

Three new *Laccaria* species from Southwest China (Yunnan)

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Abstract In this paper descriptions of three new *Laccaria* species from Southwest China (Yunnan) are reported. Macromorphological, micromorphological, and molecular data have been studied to describe the new species and delineate them within the genus *Laccaria*. The first species *Laccaria fulvogrisea* is characterized by a grey to brownish coloured fruiting body and large echinulate spores. *Laccaria aurantia* is characterized by the deeply orange colour and globose balloon-like spores with a fine echinulate ornament. The third species *Laccaria yunnanensis* has large barrel-shaped pleurocystidia and a brownish to flesh-coloured fruiting body. All three species have 4-spored basidia and were found near or within *Quercus* and *Lithocarpus* mixed broad leaved forests at an altitude of above 2,000 m.

Keywords Basidiomycota · Taxonomy · Agaricales · Hydnangiaceae · *Laccaria* · China

Introduction

Berkeley and Broome in 1883 were the first who identified the genus *Laccaria* due to the combinations of characteristics like brownish to violet colours and globose spores.

All species of the genus *Laccaria*, as well the other members of the family Hydnangiaceae, are important ectomycorrhizal fungi interacting with a wide range of host

plants all over the world. Consequently, species of the Hydnangiaceae play an important role as symbionts in forest ecosystems (McNabb 1972; Mueller 1992; Nara et al. 2003; Roy et al. 2008).

Laccaria species were identified as pioneer species at disturbed or heavy metal polluted sites. Therefore, they play a decisive role in primary and secondary succession of forest ecosystems (Trowbridge and Jumpponen 2004; Nara et al. 2003; Obase et al. 2007).

Nevertheless, the exact phylogenetic position of the Hydnangiaceae within the Agaricomycetes and the relationship between the different species within the genus *Laccaria* are not yet clarified. Its classification within the agarics was uncertain for a long time. Both morphological features (e. g., basidiospores with spines formed by perpendicular microtubules) and molecular characters segregate *Laccaria* from Tricholomataceae and group it in Hydnangiaceae (Gäumann and Dodge 1928; Besson and Kuehner 1971; Singer 1986; Mueller 1992). Only a few morphological and systematically informative characters are available, and a lot of them like the colour of the fruiting body vary in a broad range (Mueller 1992).

In recent years it could be shown that molecular and detailed morphological studies revealed that a specimen labeled with one species name encompassed hidden species with limited areas of distribution. Especially the genus *Laccaria* tends to cryptic speciation (Sheedy et al. 2013).

Members of the genus *Laccaria* are commonly found in the forests of the southwestern province Yunnan in China (Wang et al. 2004; Yang 2005). Yunnan is one of the most important biodiversity hotspots in Asia (Myers et al. 2000). More than half of the known plant species of China are present in this province (Xu and Wilkes 2004; Chaplin 2005).

According to estimates by Yang (2005), there are about 10,000 fungal species known in China. There are 40 % of them occurring in Yunnan. Three new *Laccaria* species

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from Southwestern China will be distinguished by macromorphological, micromorphological, and molecular characters.

Material and methods

Collection and field sites

The fruiting bodies were collected in the summer of 2007 in two regions within the province Yunnan, around the field station in Ailao Shan and the forests nearby Dadugang. The exact position of each collection was recorded by using a Garmin GPSmap 62 (Garmin Deutschland GmbH, Garching, Germany).

Each collection was photographed, documented, and air-dried using an electric dryer. All descriptions of macroscopic characters are based on fresh collections. The specifications of the localities and surrounding vegetation are based on transcripts of the collection sites.

Holotype collections of the new described species are deposited in the Herbarium of Cryptogams, Chinese Academy of Sciences (KUN). Isotype collections are deposited in the Herbarium Marburgense, University of Marburg, Germany.

Microscopy

Light microscopy

All examinations were observed on material mounted in 5 % KOH, water, or in a solution of lactic acid (90 %)/glycerol (10 %) to store the slides for a longer time, using a Laborlux D (Leica, Wetzlar, Germany).

Detailed measurements of at least 20 spores each, for basidia and cystidia, were taken (if possible) from at least two collections. Spore dimensions were given excluding ornamentation and the apiculus. In addition the length/wide ratio (=Q) was calculated.

Fluorescence microscopy

To investigate the number of nuclei per basidiospore, a solution of WGA-AF 488 (Invitrogen, Karlsruhe, Germany) and propidium iodine (Merck Schuchardt OHG, Hohenbrunn, Germany) was used according to the protocols from Döhlemann et al. (2011) and analysed with a Laser Scanning Spectral Confocal Microscope (CLSM, Leica, Germany).

Scanning Electron Microscopy (SEM)

For SEM studies, fragments of the lamellae of the dried material were taken, sputter coated with gold, and analysed with a Hitachi S530 (Hitachi, Japan).

DNA extraction, PCR amplification, DNA sequencing

The DNA was extracted by using the High Pure PCR Template Preparation Kit (Roche, Mannheim, Germany).

For amplifying the internal transcribed spacer (ITS) of the nuclear ribosomal DNA (rDNA) by means of a nested polymerase chain reaction (PCR), the primer combination NSII and NLB4 (Kendall and Rygielwicz 2005) was used for the first reaction. For the second specific reaction, the primers ITS1F (Gardes and Bruns 1993) and ITS4 (White et al. 1990) were utilized. The amplification with the first primer combination was carried out in a pefSTAR 96 Universal (PEQLAB Biotechnologie GmbH, Erlangen, Germany) under the following conditions: initial denaturation for 2 min at 94 °C, followed by 30 cycles with 30 s at 95 °C, 60 s at the primer-specific annealing temperature (54 °C), 180 s at 72 °C, and a final elongation step of 10 min at 72 °C.

The amplification with the second primer combination was carried out in the same thermocycler under the following conditions: initial denaturation for 3 min at 94 °C, followed by 28 cycles with 30 s at 95 °C, 60 s at the primer-specific annealing temperature (52 °C), 40 s at 72 °C, and a final elongation step of 10 min at 72 °C. The purification of the samples was performed using a PCR purification kit (Invitex, Berlin, Germany). The purified samples were sent to LGC, Berlin, Germany for sequencing.

Phylogenetic analyses

We included all sequences from the study from Wilson et al., 2013 as well as our own material and sequences from Genbank and UNITE for this study. Sequences used in this study are listed in Table 1.

The sequences were edited with CodonCode Aligner (CodonCode Aligner version 3.7.1). Other sequences of *Laccaria* species used for the analyses originated from Genbank (Benson et al. 2005) and UNITE (Abarenkov et al. 2010).

Sequence alignments were examined using Mafft (Katoh and Standley 2013). The sequence alignments were performed by using the ClustalW algorithm (FFT-NS-1 algorithm) and corrected using the Gblocks tool version 0.91b (Castresana 2000). The final dataset contained a total of 640 positions and 60 taxa.

Maximum likelihood (ML) analyses were performed with RAxML-HPG BlackBox (8.0.0) using a GTRCAT model via the Cipres Science Gateway (Stamatakis et al. 2005, 2008;

Table 1 ITS sequences from species used in this study

Order/Family	Species	Location	Voucher specimen	Genbank accession no.	Mycobank accession no.	Reference
Agaricales, Hydnangiaceae	<i>Laccaria fulvogrisea</i> spec. nov.	China, Yunnan	KUN-F 78556	JQ670896.1	MB564442	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria fulvogrisea</i> spec. nov.	China, Yunnan	MB-FB-101105	JQ681210.1	MB564442	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria aurantia</i> spec. nov.	China, Yunnan	KUN-F 78557	JQ670895.1	MB564443	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria aurantia</i> spec. nov.	China, Yunnan	MB-FB-101109	JQ681209.1	MB564443	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria yunnanensis</i> spec. nov.	China, Yunnan	KUN-F 78558	JQ670897.1	MB564444	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria yunnanensis</i> spec. nov.	China, Yunnan	MB-FB-101108	JQ681208.1	MB564444	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria proxima</i>	India	-	AJ699073.1	MB438537	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria laccata</i>	Europe	-	UDB000106	MB119173	UNITE
Agaricales, Hydnangiaceae	<i>Laccaria proxima</i>	North America	-	DQ149852.1	MB438537	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria nobilis</i>	North America	-	DQ149861.1	MB106943	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria bicolor</i>	Europe	-	UDB001494	MB332790	UNITE
Agaricales, Hydnangiaceae	<i>Laccaria bicolor</i>	China	-	JX504159.1	MB332790	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria bicolor</i>	North America	-	JX504116.1	MB332790	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria pumila</i>	North America	-	DQ149864.1	MB248633	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria montana</i>	North America	-	DQ149866.1	MB316189	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> aff. <i>montana</i>	North America	-	JX504097.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria amethystina</i>	Europe	-	AM113954.1	MB433576	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria amethysteo-occidentalis</i>	North America	-	FJ627031.1	MB114139	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria ohiensis</i>	North America	-	JX504106.1	MB287397	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504130.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504160.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria pseudomontana</i>	North America	-	DQ149871.1	MB500895	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria murina</i>	Japan	-	AB211271.1	MB271842	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria angustilamella</i>	China	-	JX504168.1	MB371125	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Japan	-	JX504083.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Japan	-	JX504084.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504163.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria acanthospora</i> TYPE	China	-	JX504102.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria alba</i>	China	-	JX504126.1	MB371126	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504123.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504128.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504127.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	India	-	JX504122.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504158.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria salmonicolor</i> TYPE	China	-	JX504143.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria himalayensis</i> TYPE	China	-	JX504101.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria bullipellis</i> TYPE	China	-	JX504100.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Japan	-	JX504081.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Japan	-	JX504089.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Japan	-	JX504091.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Japan	-	JX504090.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria ochropurpurea</i>	North America	-	JX504169.1	MB295607	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria ochropurpurea</i>	North America	-	EU819479.1	MB295607	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria tortilis</i>	Europe	-	DQ149872.1	MB438538	Genbank

Table 1 (continued)

Order/Family	Species	Location	Voucher specimen	Genbank accession no.	Mycobank accession no.	Reference
Agaricales, Hydnangiaceae	<i>Laccaria masoniae</i>	Australia	-	JX504139.1	MB530053	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> cf. <i>lateritia</i>	Australia	-	JX270719.1	MB332794	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Australia	-	JX504175.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria canaliculata</i>	Australia	-	JX504136.1	MB100701	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Australia	-	JX504174.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504125.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504134.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504135.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria vinaceoavellanea</i>	Japan	-	JN942785.1	MB316190	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Japan	-	JX504086.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Japan	-	JX504088.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Malaysia	-	JX504093.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	China	-	JX504133.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria</i> sp.	Papua New Guinea	-	JX504138.1	-	Genbank
Agaricales, Hydnangiaceae	<i>Laccaria negrimarginata</i>	China	-	JX504120.1	-	Genbank
TYPE						
Agaricales, Psathyrellaceae	<i>Mythicomycetes corneipes</i>	Germany	-	KC964108.1	MB563788	Genbank

Stamatakis 2006; Miller et al. 2010). The branches of the ML tree were tested by using a bootstrap analysis (Felsenstein 1985) with 500 replications. Bootstrap values below 60 are not shown.

Bayesian analysis was performed with MrBayes 3.2.2 (Huelsenbeck and Ronquist 2001; Ronquist and Huelsenbeck 2003; Ronquist et al. 2012) using the Cipres Science Gateway (Miller et al. 2010). Bayesian analysis included two independent runs of 10 million generations with four chains per run, a sampling frequency of 1,000 and print frequency to 1,000 generations. A burnin of 25 %, nst=1, rates=+G, and, for other parameters, default settings were used. As a result, the 50 % majority rule plus all compatible groups consensus tree from both analyses were used.

Results

Laccaria fulvogrisea F. Popa, Rexer, G. Kost, spec. nov

[KUN-F 78556] Figs. 1, 4a, b

Diagnosis Basidiomata parva ad usque media statura; pileus maxime 30 mm latus, convex ad usque plano-convex, glabrus, griseus; margine pilei tenuie et translucente striato; lamellae distantes, ventricosae, adnexae ad usque adnatae, furcatae et 5 mm latae, color basidiomata iuveniliū subalbidus, in senectute griseus ad brunneus; stipes 30–70 x 3–5 mm,

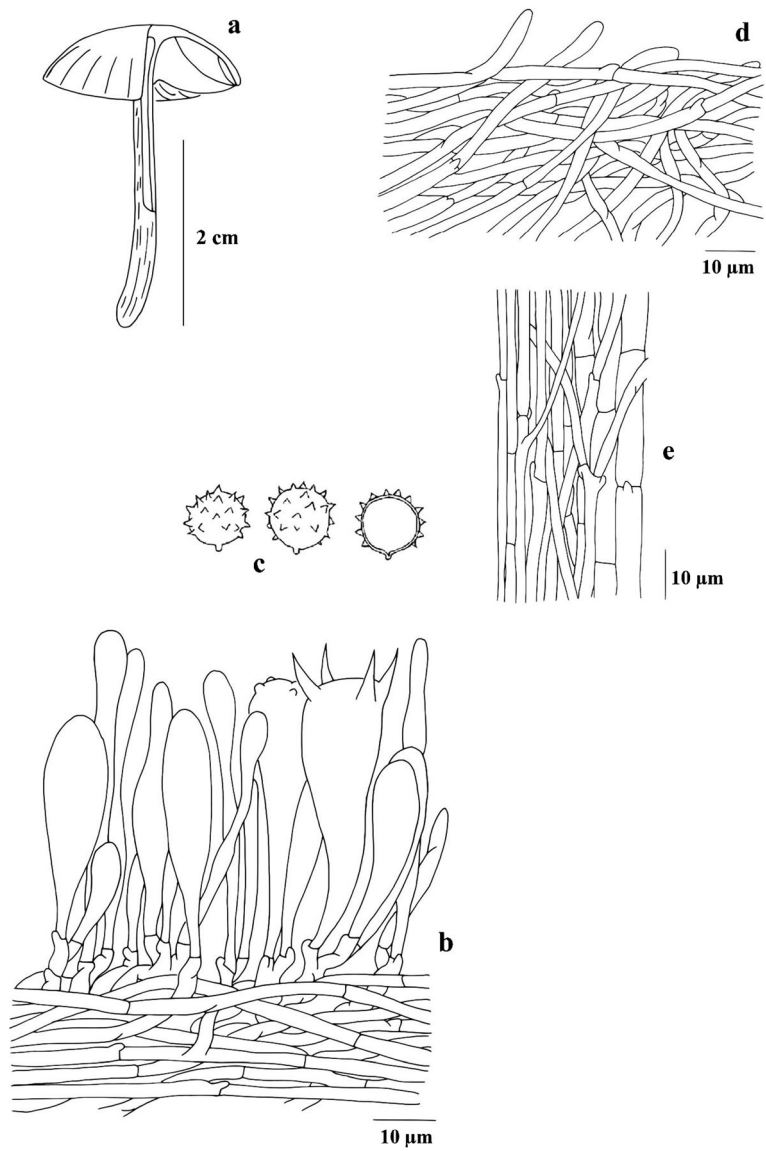
cylindricus, fistulatus, glabrus to molliter fulgens, griseus paulum tenebrosior quam pileus; basidiomata iuveniles cum colore brunneo intermixto; sine caulocystidiis; mycelium basale albo colore; hyphae pileipellis semper hyphis intertextae et ascendentes perpendiculis clavatis; basidia (38.5) 40–50 (55) x (8.8) 10–15 (16.5) μ m, clavata, hyalina, 4-sporigera; sterigmata 10 μ m longa; basidiosporae mensio sporarum sine echinulis (7.5–10) 8–10 x 8–11 (7.5–11) μ m, Q=1–1.1 (1.25), globosae ad usque subglobosae, hyalinae, non-amyloideae, echinatae, echinulae (1.5) 1.7–2.5 (2.55) μ m longae, hilar-appendices 2 μ m, duo nuclei in basidiospora; pleurocystidia clavata, flexuosa; cheilocystidia clavata etiam subcapitata (25) 30–50 x 3–7 (8) μ m; omnes hyphae fibulatae.

Holotypus K. Donges, G. Kost, Rexer [KUN-78556], 16. VII. 2007, Ailao Shan (N24 32.870 E101 01.966, altitude 2,496 m), Yunnan, China. Mycobank accession number: MB564442; GenBank accession number: JQ670896

Basidiomata (Fig. 1) scattered, small to medium sized. Pileus <30 mm in diam., initially convex to plano-convex, flattening with age, sometimes becoming centrally depressed, glabrous, translucent-striate, grey and with a violet to reddish brown tinge. Flesh thin, at margin very thin. Lamellae distant, ventricose, up to 5 mm broad, adnexed to adnate, furcate, initially whitish, greyish to brownish with age, and dried $\pm$ brownish edge.

Stipe 30–70 x 3–5 mm, cylindrical, fistulose, glabrous to softly gleaming, grey, slightly darker than the pileus, in young

Fig. 1 *Laccaria fulvogrisea* [KUN-F 78556] **a.** Basidiomata **b.** Hymenium **c.** Spores **d.** Pileipellis **e.** Caulopellis



fruiting bodies with brownish hue, and base covered with a white mycelium.

Basidia (38.5) 40–50 (55) x (8.8) 10–15 (16.5) μm , clavate, hyaline, 4-spored, and sterigmata up to 10 μm long. Basidiospores (7.5) 8–10 x 8–11 (7.5) μm , globose to subglobose, $Q=1\text{--}1.1$ (1.25), hyaline, echinulate, echinulae (1.5) 1.7–2.5 μm , hilar appendix 2 μm long, and two nuclei per basidiospore. Pleurocystidia clavate to flexuose, cheilocystidia clavate-subcapitate (25) 30–50 x 3–7 (8) μm . Clamp connections present. Pileipellis consisting of interwoven radiating hyphae with a few perpendicular clavate hyphal ends.

Etymology Named after the grey fruiting body with a brownish tint.

Geographic distribution/host range Southwestern China in the Ailao Shan Mountains, Yunnan at approximately 2,500 m altitude. Locality: between mosses on soil within a mixed broadleaf forest with *Quercus* spp., *Eucalyptus* spp., *Lithocarpus* spp. and *Ficus* spp., in the understorey where mainly bamboo dominates. The sequences from this species show a great similarity to uncultured *Laccaria* species (GenBank: FJ827221.1), which are in an ectomycorrhizal association with *Polygonum* and *Potentilla* species on an alpine meadow in southwestern China (Gao and Yang 2009, unpublished).

Material examined China, Yunnan: Ailao Shan (N24 32.870 E101 01.966, alt. 2,496 m), 16. VII. 2007, K. Donges, G. Kost, Rexer [KUN-F 78556, *holotype*]; China, Yunnan: Ailao Shan (N24 32.741 E101 01.479, alt. 2,474 m), 14. VII. 2007, K. Donges, G. Kost, Rexer [MB-101105]; China, Yunnan: Ailao Shan (N24 32.735 E101 01.481, alt. 2,497 m), 16. VII. 2007, K. Donges, G. Kost, Rexer [MB-101102]; China, Yunnan: Ailao Shan (N24 32.735 E101 01.481, alt. 2,497 m), 16. VII. 2007, K. Donges, G. Kost, Rexer [MB-101103]; China, Yunnan: Ailao Shan (N24 32.929 E101 01.403, alt. 2,488 m), 16. VII. 2007, K. Donges, G. Kost, Rexer [MB-101104]; China, Yunnan: Ailao Shan (N24 32.728 E101 01.471, alt. 2,491 m), 14. VII. 2007, K. Donges, G. Kost, Rexer [MB-101110].

Commentary *Laccaria fulvogrisea* can be characterized by its grey to brownish coloured basidiomata, 4-spored basidia, and globose spores with extensive echinulation.

Laccaria fulvogrisea shows a violet hue like *Laccaria amethystina* Cooke, but in *Laccaria amethystina* the grey to brownish colours are lacking. To a large extent *Laccaria fulvogrisea* is similar to the grey to dark grey *Laccaria murina* S. Imai, described from Japan (Imai 1938), and also to *Laccaria nigra* Hongo, but in both species brown and violet tinges are lacking and *Laccaria nigra* has 2-spored basidia.

Both *Laccaria violaceonigra* G. Stev. and *Laccaria olivaceogrisea* Vellinga show dark grey colours, but differ especially in the shape of the spores and distinctive colour combinations.

In addition, the spores of *Laccaria fulvogrisea* have stronger echinulation than the spores of the discussed species.

Laccaria aurantia F. Popa, Rexer, G. Kost, spec. nov

[KUN-F 78557] Figs. 2, 4b

Diagnosis Basidiomata parva ad usque media statura; pileus maxime 35–40 mm latus, convex vel applanatus, glabrus et aurantius, basidiomata aurantia; lamellae distantes, ventricosae ad usque subventricosae, adnatae, 5 mm latae; stipes 80 mm, cylindraceus, fistulatus, aurantius cum colore brunneo intermixto; sine caulocystidiis; mycelium basale albo colore; pileipellis semper hyphis intertextis, clavatae hyphae sine perpendiculis, paries non crassa; basidia (38–50) 40–50 x (9–13) 10–12 μm , clavata, hyalina, 4-sporigera, interdum bispora; sterigmata 10 μm longa; basidiosporae mensio sporarum sine echinulis (8–11) 9–10 x 8–10 (7–10) μm , $Q=1\text{--}1.05$ (1.57), globosae ad usque subglobosae, hyalinae, non-amyloideae, echinatae, echinulae (0.75) 1.1–1.5 (1.75) μm longae, hilar-appendix 1–1.5 μm , duo nuclei in basidiospora; pleurocystidia et cheilocystidia similia (20–45) 25–40 x 5–10 (3–11) μm clavata et flexuosa; omnes hyphae fibulatae.

Holotypus K. Donges, G. Kost, Rexer [KUN-78557], 16. VII. 2007, Ailao Shan (N24 32.726 E101 01.732, alt. 2,484 m), Yunnan, China. Mycobank accession number: MB564443; GenBank accession number: JQ670895

Basidiomata (Fig. 2) scattered to gregarious, medium sized. Pileus 35–40 mm in dia., hemispherical to convex, flattened with age, centrally depressed to umbilicate, glabrous, $\pm$ smooth, not or only slightly translucent-striate, hygrophan, and vividly orange coloured. Lamellae distant, adnate, ventricose to broadly ventricose, up to 5 mm broad, smooth, orange coloured, and with edge concolorous.

Stipe up to 80 mm long, cylindrical, centrally inserted, fistulose, orange with brownish coloured covering, and with a whitish basal mycelium.

Basidia 40–45 (50) x 10–12 (13) μm , clavate, hyaline, 4-spored, some 2-spored, sterigmata up to 10 μm long. Basidiospores (8) 9–10 (11) x (7) 8–10 μm , globose to subglobose, $Q=1.05$ (1.57), hyaline, echinulate, echinulae (0.75) 1–1.5 (1.75) μm , hilar appendix 1–1.5 μm long, and two nuclei per basidiospore. Pleurocystidia and cheilocystidia (20) 25–40 (45) x (3) 5–10 μm , cylindrical to flexuous. Clamp connections

Fig. 2 Basidiomata of *Laccaria aurantia* [KUN-F 78557] (scale 2 cm). *Laccaria aurantia* [KUN-F 78557] **a.** Basidiomata **b.** Hymenium **c.** Spores **d.** Pileipellis **e.** Caulopellis

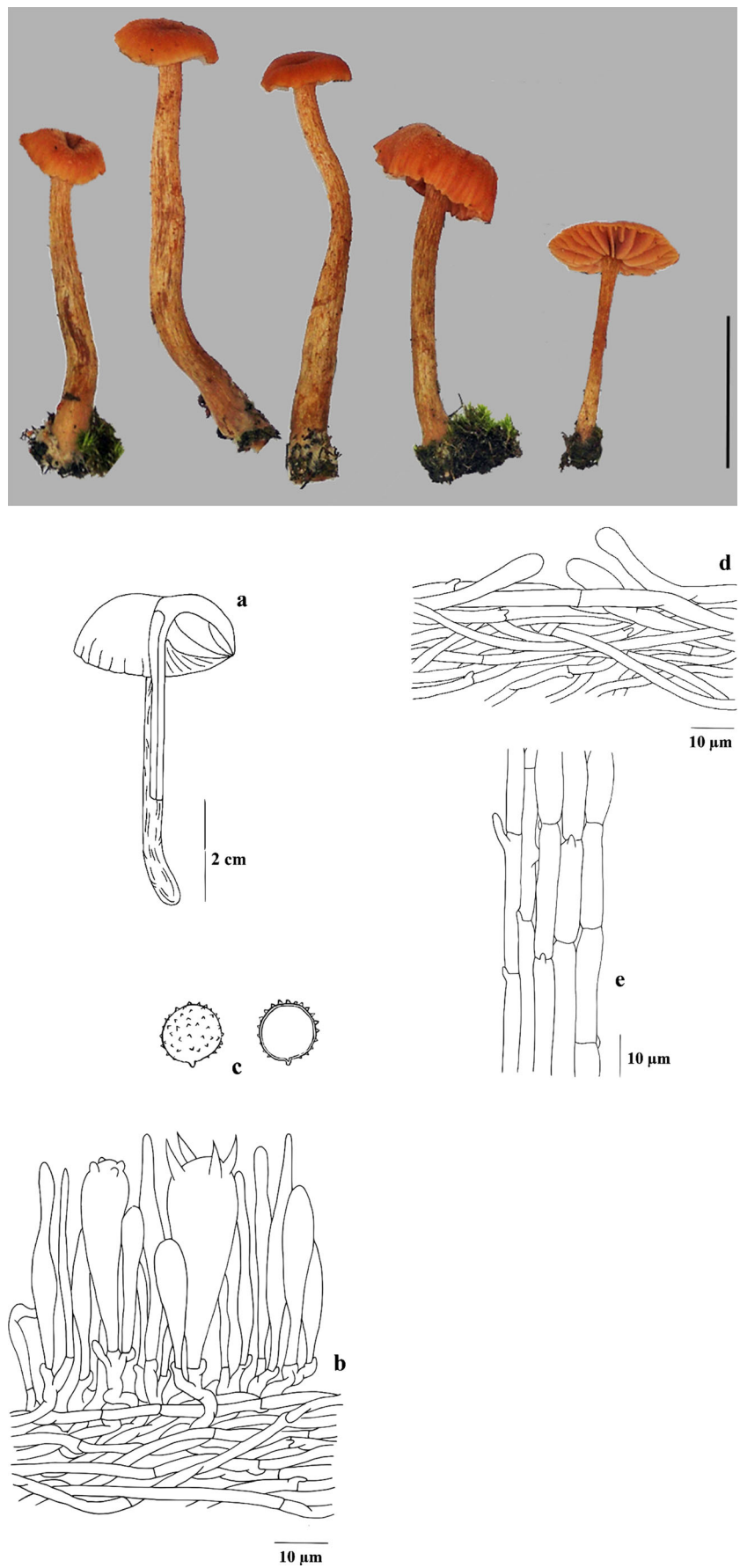


Fig. 3 Basidiomata of *Laccaria yunnanensis* [KUN-F 78558] (scale 2 cm). *Laccaria yunnanensis* [KUN-F 78558] **a.** Basidiomata **b.** Hymenium **c.** Spores **d.** Pileipellis **e.** Caulopellis

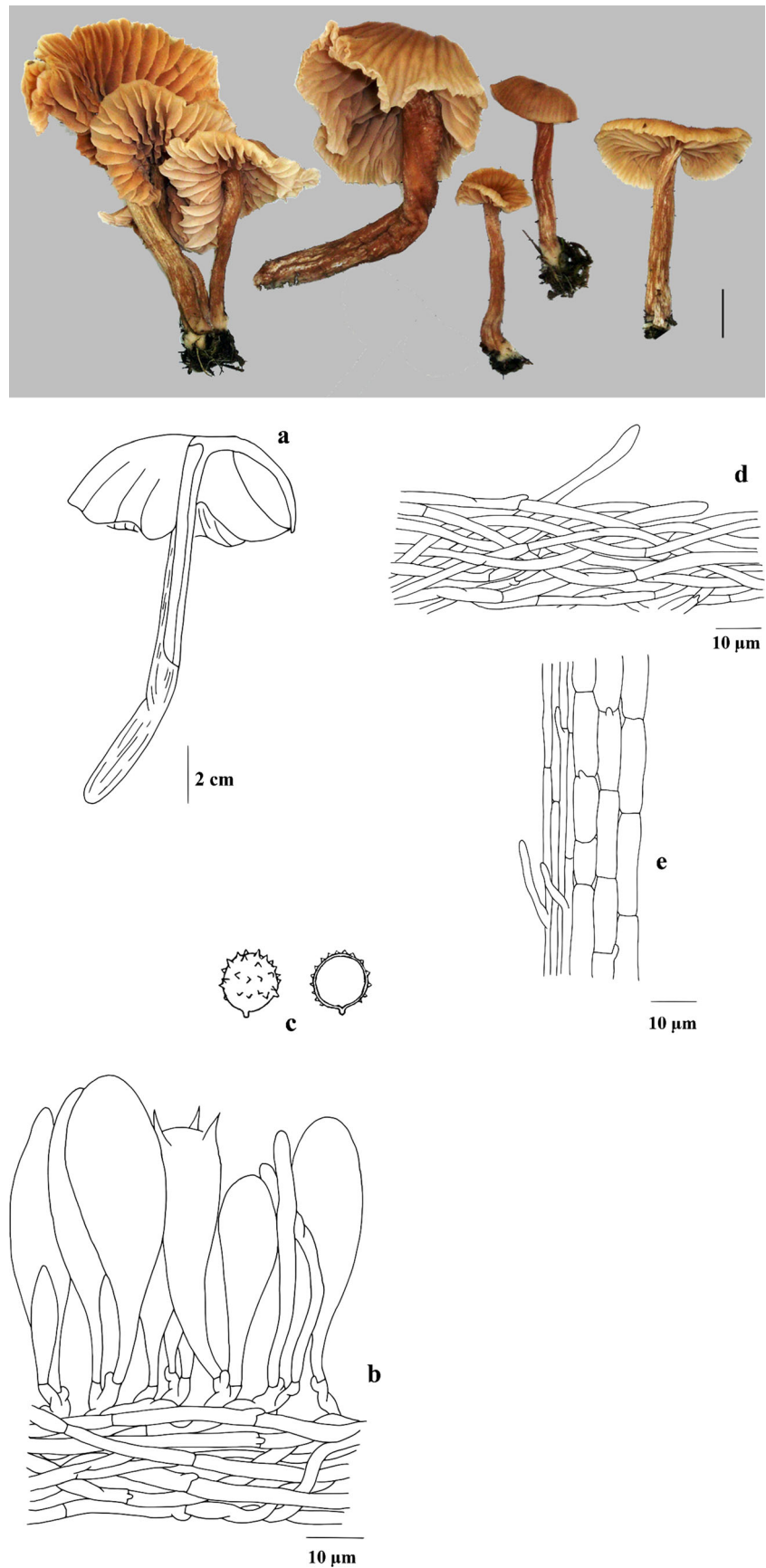
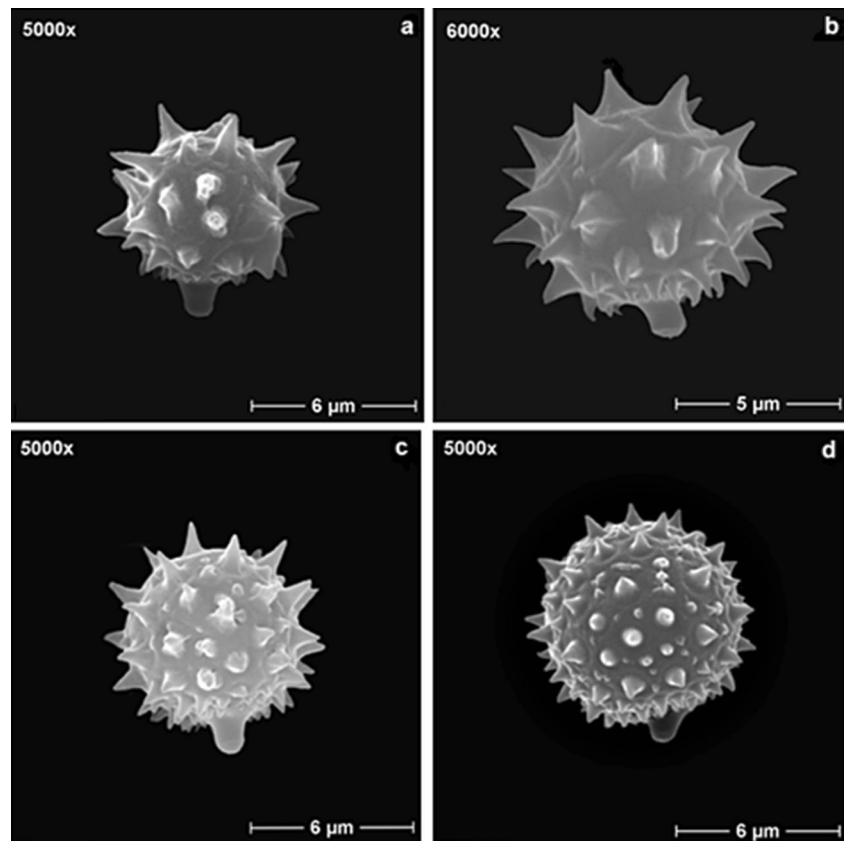


Fig. 4 SEM's of basidiospores: **a.** *Laccaria fulvogrisea* [KUN-F 78556] (young Spore) **b.** *Laccaria fulvogrisea* [KUN-F 78556] (mature Spore, other collection) **c.** *Laccaria yunnanensis* [KUN-F 78558] **d.** *Laccaria aurantia* [KUN-F 78557]



present. Pileipellis consisting of interwoven radiating thin-walled hyphae, without perpendicular hyphal ends.

Etymology This species is named because of the conspicuous orange colour of the fruiting body.

Geographic distribution/host range Southwestern China in the Ailao Shan Mountains (N24 32.726 E101 01.732), Yunnan at approximately 2,500 m altitude. It was collected within a mixed broadleaf forest dominated by *Quercus* spp. and *Lithocarpus* spp.

Material examined China, Yunnan: Ailao Shan (N24 32.726 E101 01.732, alt. 2,484 m), 16. VII. 2007, K. Donges, G. Kost, Rexer [KUN-F 78557, *holotype*]; China, Yunnan: Ailao Shan (N24 32.602 E101 01.641, alt. 2,478 m), 17. VII. 2007, K. Donges, G. Kost, Rexer [MB-101109].

Commentary *Laccaria aurantia* is characterized by its vividly orange coloured basidiomata, the globose, balloon-like spores with many short spines, orange lamellae and 4- or 2-spored basidia. This species is most similar to *Laccaria laccata*, but also to *Laccaria glabripes* and *Laccaria echinospora* (McNabb 1972), both brownish species are known from New Zealand. *Laccaria echinospora* differs from *Laccaria aurantia*

having 2-spored basidia and *Laccaria glabripes* having longer spines.

Laccaria aurantia can also be distinguished by the thin, flesh pink to reddish brown sulcate pileus and a glabrous stipe with a cartilaginous appearance.

Laccaria yunnanensis F. Popa, Rexer, G. Kost, spec. nov

[KUN-F 78558] cprFigs. 3, 4c

Diagnosis Statura basidiomatum media; pileus maxime 60–100 mm latus, convex ad hemisphaericus, glabrus; basidiomata ferruginea et incarnata, margine pilei tenuie et translucente striato; lamellae distantes ventricosae, adnexae ad usque adnatae, in altum 5 mm; stipes 30–100 mm, cylindricus, distortus, fistulatus, tenebrosior quam pileus; sine caulocystidiis; mycelium basale albo colore; pileipellis semper hyphis intertextis, clavatae hyphae sine perpendicularis, paries non crassa; basidia (40–50) 45–50 x 9–10 (9–11) µm, clavata, hyalina, 4-sporigera, sterigmata 8 µm longa; basidiosporae mensio sporarum sine echinulis (7.5–9) 8–9 x 8–10 (7.5–10) µm, Q=1.05 (1.25), globosae ad usque subglobosae, hyalinae, non-amyloideae, echinatae, echinulae (0.75) 1–1.5 (1.75) µm longae, hilar-appendix 1–1.5 µm, duo nuclei in basidiospora; pleurocystidia aliquantus, clavata vel

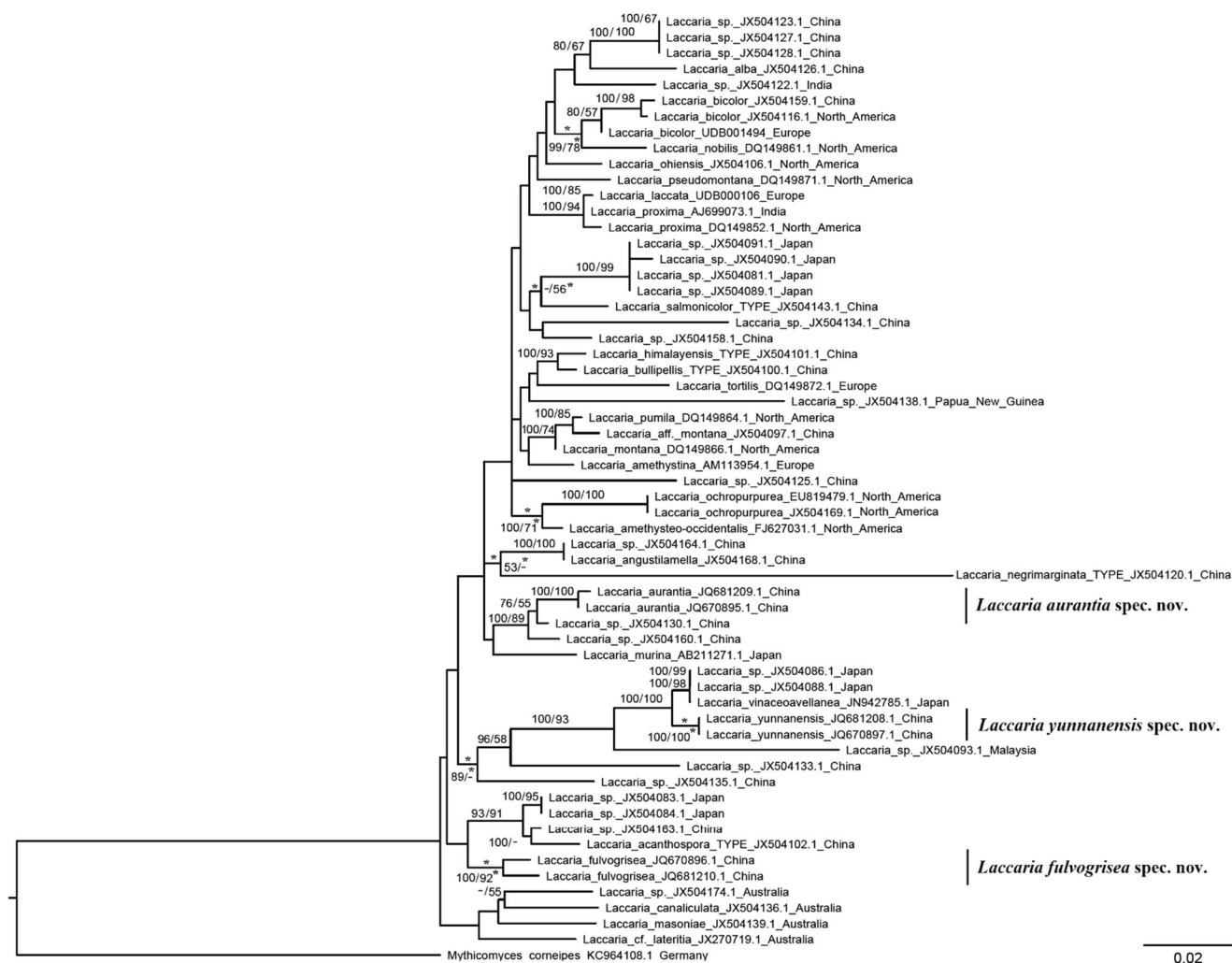


Fig. 5 Maximum-likelihood (ML) tree based on ITS 1, 5.8S and ITS 2 sequences of three new Chinese *Laccaria* species and other *Laccaria* sequences from GenBank and UNITE. The right number denotes the bootstrap value (500 replicates). The left number indicates the Bayesian analysis probability values in percent

ellipsoidea, cylindrica vel conica (50–70) 55–65 x 15–25 (10–30) μm ; cheilocystidia (25–55) 30–50 x 4–6 (3–7) μm clavata vel flexuosae; omnes hyphae fibulatae.

Holotypus K. Donges, G. Kost, Rexer [KUN-78558], 16. VII. 2007, Ailao Shan (N24 32.726 E101 01.732, alt. 2,484 m), Yunnan, China. Mycobank accession number: MB564444; GenBank accession number: JQ670897

Basidiomata (Fig. 3) scattered to gregarious, medium sized to large. Pileus 60–100 mm in dia., convex to hemispherical, soon flattened, concave to infundibuliform with age, glabrous, brownish to flesh-coloured, margin very thin, translucent-striate and straight, undulate to eroded and reflexed with age. Lamellae distant, ventricose to broadly ventricose, adnexed to adnate and up to 5 mm broad, flesh-coloured. Stipe 30–100 x 3–15 mm, cylindrical, centrally inserted, distorted, fistulose, glabrous, darker than the pileus, and mycelium at the base of the stipe whitish.

Basidia (40) 45–50 x 9–10 (11) μm , clavate, hyaline, 4-spored, sterigmata up to 8 μm long. Basidiospores excluding ornamentation (7.5) 8–9 x (7.5) 8–10 μm , $Q=1.05$ (1.25), globose to subglobose, hyaline, echinulate, echinulae (0.75) 1–1.5 (1.75) μm with a 1–1.5 μm long hilar appendix and two nuclei per basidiospore. Pleurocystidia (50) 55–65 (70) x (10) 15–25 (30) μm , clavate to ellipsoid, cheilocystidia (25) 30–50 (55) x (3) 4–6 (7) μm , cylindrical sometimes conical, and clavate to flexuous. Clamp connections present. Pileipellis consisting of radiating interwoven thin-walled hyphae.

Etymology *L. yunnanensis* is named after the collecting locality.

Geographic distribution/host range It was found nearby *Laccaria fulvogrisea* in southwestern China in the

Ailao Shan Mountains (N24 32.726 E101 01.732), Yunnan at approximately 2,500 m altitude. The holotype found between grasses on a disturbed site, and the second collection on an open ground in a ditch beside a gravel forest road.

Material examined China, Yunnan: Ailao Shan (N24 32.726 E101 01.732, alt. 2,484 m), 16. VII. 2007, K. Donges, G. Kost, Rexer [KUN-F 78558, *holotype*]; China, Yunnan: Dadugang (N22 22.452 E100 55.816, alt. 1,398 m), 23. VII. 2007, K. Donges, G. Kost, Rexer [MB-101108].

Commentary *Laccaria yunnanensis* is characterized by its brownish to flesh-coloured basidioma, convex to hemispherical large striate pileus, 4-spored basidia and large pleurocystidia.

The other species within this genus differ from *Laccaria yunnanensis* by missing the large pleurocystidia and very striking striate pileus (Singer 1977; Contu 1998, 2003).

Laccaria vinaceoavellanea (Hongo 1971) is very similar to *Laccaria yunnanensis*, but the typical pleurocystidia are lacking. Molecular data (Fig. 5) provide evidence for the separation of and the close relationship between both species.

Discussion

The number of *Laccaria* species described from Europe is still increasing. Since 1990, four species were described as new to science (Contu 1998, 2003; Fernández & Contu 2003; Pázmány 1994). In the same period, only two species were described from Yunnan (Wang et al. 2004). This does not mean there is nothing to describe in that region because, the region of Yunnan is known to be one of the most important biodiversity hotspots of the world. Yunnan shows a broad range of different ecosystem types from tropical broadleaved rain forests in the south of the province to high montane coniferous forests in the north. Influences of Southeast Asian as well as northern temperate regions additionally increase the biodiversity of Yunnan (Xiwen et al. 1986; Mueller et al. 2001). *Laccaria* species are widely distributed and very common throughout various ecosystem types in Yunnan (Wang et al. 2004). A huge number of undescribed species can be expected from this region.

The ectomycorrhizal genus *Laccaria* is well known for species such as *Laccaria amethystina*, which could be shown to be a real generalist, interacting with a wide range of host trees and being located in various habitat types (Vincenot et al. 2011). Nevertheless, there are species like *Laccaria maritima*, which is restricted to occur only in sand dune habitats in northern Europe and Greenland (Mueller 1992). In Yunnan on the one hand we found species, which were similar to European species (data not shown). This might support the

opinion that many of the well-known *Laccaria* species, like *Laccaria laccata* and *Laccaria amethystina* were found worldwide across all ecosystem types. However, in recent years molecular and detailed morphological studies revealed that specimen labeled with one species name often encompass hidden species with limited areas of distribution (Taylor et al. 2006; Jargeat et al. 2010; Vincenot et al. 2011). On the other hand, we had collections of species, which were only found in one habitat type. Some of these collections were described in this paper as new for science.

By means of ITS-sequences (Fig. 5), morphology as well as ecology of all three new species could be clearly separated from the other known and recently described *Laccaria* species (Wang et al. 2004; Wilson et al. 2013). The morphological similarities of *Laccaria yunnanensis* and *Laccaria vinaceoavellanea* were confirmed by the molecular analysis, and so a close relationship between both species is well supported (Fig. 5).

Our phylogenetic analysis is not supporting the established taxonomical arrangement of the genus *Laccaria*. For a better insight for that purpose, a much broader dataset is needed. However, it is obvious that the basal groups in our phylogenetic tree (Fig. 5) are originating from Australia and New Zealand. Taking into account the variation of colour of the *Laccaria* taxa, Vellinga (1986) hypothesized that the genus *Laccaria* originated from those regions.

There were no other phylogeographic relationships detectable from our phylogenetic studies. Consequently, for a better understanding of the phylogenetic relationships of the species within the genus *Laccaria*, additional data were needed. These further investigations should concentrate on Southeast Asia and Australia. It seems as if in this region a hot spot of biodiversity of the genus *Laccaria* with diverse phylogeographic relations is located.

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